13 A Framework for Prediction

How to carry out a prediction, and how to find the best predicting model

Motivation

You have a car that you want to sell in the near future. You want to know what price you can expect if you were to sell it. You may also want to know what you could expect if you were to wait one more year and sell your car then. You have data on used cars with their age and other features, and you can predict price with several kinds of regression models with different right-hand-side variables in different functional forms. How should you select the regression model that would give the best prediction?

You want to predict sales at an ice cream shop for the next few days. You have data from the recent past on daily sales and temperature, and you have good weather forecasts for the next few days. How should you use this data to make sales predictions that will be as close to actual sales as possible? What issues should you consider to make sure that the prediction from your data will be as good as possible for the future situation you are interested in?

Part III of our textbook introduces the logic of predictive data analysis and its most widely used methods. We focus on predicting a target variable y with the help of predictor variables x. The basic logic of prediction is estimating a model for the patterns of association between y and x in existing data (the original data), and then using that model to predict y for observations in the prediction situation (in the live data), in which we observe x but not y. The task of predictive analytics is to find the model that would give the best prediction in the live data by using information from the original data.

This chapter introduces a framework for prediction. We discuss the distinction between various types of prediction, such as quantitative predictions, probability predictions, and classification, and we focus on the first of these. We introduce point prediction versus interval prediction; we discuss the components of prediction error and how to find the best prediction model that will likely produce the best fit (smallest prediction error) in the live data, using observations in the original data. We introduce loss functions in general, and mean squared error (MSE) and its square root (RMSE) in particular, to evaluate predictions. We discuss three ways of finding the best predictor model: using all data and the Bayesian Information Criterion (BIC) as the measure of fit, using training–test splitting of the data, and using k-fold cross-validation, which is an improvement on the training–test split. We discuss how to assess and, if possible, improve the external validity of predictions. We close the chapter by discussing what machine learning means.

The case study in this chapter is **Predicting used car value with linear regressions**, which uses the `used-cars` dataset on used cars of a specific model advertised in a single city. We use this case study to illustrate the main issues we need to address in predictive data analysis and the logic of model selection.

Learning outcomes

After working through this chapter, you should be able to:

- identify situations where you want to, and can, predict y with the help of x;
- understand the concept of prediction error and its components;
- carry out point predictions and interval predictions of quantitative outcomes with linear regression;
- understand the role of model complexity in overfitting the original data;
- use BIC and k-fold cross-validation to find the model that best fits the population, or general pattern, behind the original data;
- assess external validity with the help of domain knowledge, and, if possible, with additional analysis of the original data.

13.1 Prediction Basics

In data analysis, prediction means assigning a value to y, the **target variable**, or **outcome variable**, for a **target observation** or more **target observations**. The value of y is not known for the target observations, but the value of one or more x variables is known. Those x variables are called **predictor variables** in this context. Data analysis with the aim of prediction is called **predictive data analysis** or **predictive analytics**.

The basic logic of prediction is as follows. We have data with observations on both y and x. We call this data the **original data**. The original data doesn't include the target observations for which we want to make the prediction. We use the original data to uncover patterns of association between y and x that we'll use in our prediction. To uncover those patterns, we specify and estimate a model. With the help of that model, we then predict the value of y for the target observations, for which we can observe x but not y. We call data that includes the target observations the **live data**. Thus, in short, we uncover patterns in the original data that we use for prediction in the live data.

Mathematically, those patterns of association are expressed as a function. Recall that a function is a rule that gives a value for y if we plug in values for x. For a target observation j in the live data, we observe the values of the predictor variables x_j. If we know the function, all we need to do is feed those x_j values into it. The abstract notation for the variable and observation we want to predict is y_j. The specific predicted value we call $\hat{y}_j$. The abstract notation for the function we use for the prediction is f; the notation for the specific function we use is $\hat{f}$:

$$\hat{y}_j = \hat{f}(x_j) \tag{13.1}$$

In data analysis, such functions are called **predictive models**. Typically, we can have many different predictive models that give some value of y if we feed in values of x, but we need the model that gives the best prediction. To be more precise, we use the original data to find the model that will then give the best prediction in the live data.

An example for a predictive model is the linear regression with a specific choice of functional forms (e.g., with interactions or without interactions). In the abstract, the linear regression with y, x_1, x_2, is a model for the conditional expected value of y, and it has coefficients β. We need estimated coefficients ($\hat{\beta}$) and actual x values (x_j) to predict an actual value $\hat{y}$:

$$y^E = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \tag{13.2}$$

$$\hat{y}_j = \hat{\beta}_0 + \hat{\beta}_1 x_{1j} + \hat{\beta}_2 x_{2j} + \cdots \tag{13.3}$$

Here the function, or model, is $f(x_1, x_2, \ldots) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots$, and its estimated version, with specific values for its coefficients, is $\hat{f}(x_{1j}, x_{2j}, \ldots) = \hat{\beta}_0 + \hat{\beta}_1 x_{1j} + \hat{\beta}_2 x_{2j} + \cdots$.

For example, we may want to predict sales of ice cream in our ice cream shop for a specific summer day. We know what day of the week that is, what month of the year, and we have a good forecast for the highest temperature on that day: Wednesday, August, and 30 degrees Celsius (88 degrees Fahrenheit). y is sales, j is the target day, the x are binary variables for day of the week and the month of the year, and temperature. We have access to daily data from three years with the sales, temperature, day of the week, and month variables. That's our original data.

One way to carry out the prediction is the following. We estimate a linear regression using all observations in our original data. That gives us the $\hat{\beta}$ coefficient values. Then we plug in the x_j values and multiply them by the estimated $\hat{\beta}$ coefficient values. The results are our $\hat{y}_j$.

In practice, we'll carry out additional steps when making a prediction. The reason is that there may be other possible models using the same data, and some of those other models could give better predictions. For example, we could have a linear regression with piecewise linear spline for temperature, or we could include interactions of temperature and day of the week, and so on. Or, we may have a model that is not a linear regression at all – we'll introduce such models in subsequent chapters. Some of those models may be better at predicting y using the same data, others less good. Finding the best of such models accounts for many additional steps in predictive analytics.

The fundamental task of predictive analytics is finding the best model. We have the original data that we can use for that. But the best model is best in the sense that it gives the best prediction in the live data, not in the original data. What exactly we mean by the best model, and whether and how we can find it using the original data, are difficult questions.

Review Box 13.1 The logic of prediction

- Prediction means finding a value for y with the help of x for observations with observed x but unobserved y.
- Data analysts use patterns of association between y and x in their original data to predict y in the live data.
- Patterns of association between y and x are captured by a model.
- The fundamental task of prediction is to find the model using the original data that gives the best prediction in the live data.

13.2 Various Kinds of Prediction

Before we move on to finding the best model, let's introduce some more concepts that help describe what kind of prediction we want.

Sometimes we want to predict the value of a quantitative outcome y: that would be a **quantitative prediction**. Sometimes our y is binary, and we want to predict the probability of $y = 1$; that would be a **probability prediction**. Yet other times, when we have a binary y, we want to predict whether the outcome is 0 or 1, not the probability. That's called **classification**. In principle, probability predictions and classifications can be applied not only to binary y variables but to qualitative y variables with more categories, too.

Our example of predicting sales in an ice cream shop is a quantitative prediction. An example for a probability prediction is the probability of whether a loan applicant will repay their loan. Here we have a binary y variable (1 if repays, 0 if defaults); the prediction would be based on the probability that an applicant would repay their loan as a function of various predictor variables that we know about the applicant when they apply for the loan. For a classification example, consider the same question of whether a loan applicant would repay their loan, but here we want to classify each applicant into one of two groups: whether they'll repay their loan or not.

With quantitative predictions, and with probability predictions, we can predict a single value $\hat{y}_j$. That is called **point prediction**, to distinguish it from **interval prediction**. An interval prediction produces a **prediction interval**, or **PI**, that tells where to find the value of y_j with a certain likelihood (e.g., a 95% prediction interval or an 80% prediction interval).

In the context of the simple linear regression, we have already introduced the prediction interval, in Chapter 9, Section 9.3: the point prediction plus or minus the standard prediction error (SPE), multiplied by a number. That multiplying number reflects the likelihood that we want for the prediction interval for covering the actual values. For a 95% PI this number is 2: $95\% PI(\hat{y}_j) = \hat{y}_j \pm 2SPE(\hat{y}_j)$; for an 80% PI it would be 1.3: $80\% PI(\hat{y}_j) = \hat{y}_j \pm 1.3SPE(\hat{y}_j)$; for a 67% PI it would be 1: $67\% PI(\hat{y}_j) = \hat{y}_j \pm SPE(\hat{y}_j)$. Recall the formula for the standard prediction error in a linear regression with a single x variable is:

$$SPE(\hat{y}_j) = Std[e]\sqrt{1 + \frac{1}{n} + \frac{(x_j - \bar{x})^2}{nVar[x]}} \tag{13.4}$$

With more x variables the formula is more complicated, but the intuition is similar. The SPE is smaller the larger the sample, the smaller the residual standard deviation, the closer the observation to the average of the x variable(s), and the larger the spread of the x variable(s). In subsequent chapters we'll consider alternative models for prediction. The concept of prediction interval, and the intuition behind its components, will be similar for those alternative models, too.

For example, a point prediction of ice cream sales is a single number, such as 1462 dollars. An example for an interval prediction would be $[1299; 1625]$ with an 80% likelihood. When we want to predict the probability of whether a loan applicant will repay their loan, a point prediction may be 0.82. We may have a corresponding interval prediction, say $[0.74; 0.90]$ with an 80% likelihood.

In this chapter and the following few chapters, we'll focus on quantitative predictions. We'll discuss probability prediction and classification in Chapter 17.

Finally, we will use the term **forecasting** for making predictions about future values of y when the original data is a time series. True, most of the time the original data is from the past and the live data, with the target observation(s), is in the future. So most predictions are about future values, and thus distinguishing forecasting from other kinds of predictions is a little arbitrary. For historical reasons, in business and economics, the term forecasting is used when the original data is a time series itself. We'll discuss forecasting in Chapter 18.

Review Box 13.2 Types of predictions

- Quantitative predictions predict the value of a quantitative y; probability predictions predict the probability of $y = 1$ for a binary variable; classifications predict whether $y = 1$ or $y = 0$ for a binary variable.
- Quantitative predictions provide a single value (point prediction) or a range of values (interval prediction).
- Forecasts are predictions that use time series data.

13.A1 CASE STUDY – Predicting Used Car Value with Linear Regressions

Question and data

The goal of this case study is to predict the price for cars with particular features. Suppose that we have a car that we want to sell in the near future. It is a regular gas-fueled Toyota Camry LE sedan that is 10 years old with 120 000 miles on it and a 4-cylinder engine, and our car is in an excellent condition. We live in the Chicago area in the United States, so we will look at the Chicago used car market. We want to predict the future selling price of the car. To make this prediction, we use data on ad prices of similar cars. Our main interest is in predicting the price we can expect to get for our car. But we may also be interested in how we can expect the price to change if we were to wait one more year, which would increase age and miles.

Predicting the price of a used car is a quantitative prediction exercise. The goal is to assign a price (point prediction) or a range of prices to cars with certain features.

In this case study, we use the `used-cars` dataset, which we collected in early 2018 from offers in Chicago. To match the car of interest, our data is on a single model, the Toyota Camry. The features include the more specific type of the car (LE, Hybrid), when it was made, how many miles it has, whether it's in good condition, and so on.

First, as usual, we started with data cleaning. We identified variable values in the data that are obvious errors and dropped those observations. Second, we created a data table with cars that are not fundamentally different from the car whose price we want to predict. We dropped cars that are found erroneously in the data (e.g. trucks) as well as hybrid cars. Our variables are price, age, miles, type, whether in good condition, whether 4 cylinders or 6 cylinders, and whether advertised by the owner or a dealer.

13.3 The Prediction Error and Its Components

The goal of a prediction is getting as close to the value of y for the target observation as possible. With a quantitative prediction, the point prediction is $\hat{y}_j$. The difference between the two is the **prediction error**:

$$e_j = \hat{y}_j - y_j \tag{13.5}$$

The actual value, y_j, is not known when the prediction is made; hence the need to predict it. Thus, the prediction error is not known, either, until we observe y_j itself. But arriving at a good prediction requires thinking about the prediction error before we get to observe it.

The ideal prediction error, e_j, is zero: our predicted value is right on target. The prediction error is positive if we over-predict the value: we predict a higher value than the actual value. It is negative if we under-predict the value: our prediction is too low. The prediction error is larger in absolute value the further away our prediction is from the actual value. It is smaller the closer we are. Whether positive versus negative errors matter more, or they are equally bad, depends on the decision problem for which we are using the prediction. We shall return to this question in the next section.

Our definition is valid for quantitative predictions. We can define the prediction error for probability predictions in a very similar way, too. For classification, the prediction error still makes sense, but it can be of few values only (classified right or classified wrong). As indicated earlier, we'll focus on quantitative predictions in this chapter.

Often we want to make predictions for more than one target observation. Then the goal is to arrive at predicted values with as small an error as possible for all of them. We shall return later to the question of how we should combine prediction errors for many observations.

The prediction error can be decomposed into three components: **estimation error**, **model error**, and **irreducible error** – also called **idiosyncratic error** or **genuine error**.

The estimation error comes from the fact that, with a model f, we use our original data to find $\hat{f}$. When using a linear regression for prediction, this error is due to the fact that we don't know the values of the coefficients of the regression (β), only their estimated values ($\hat{\beta}$). This error is captured by the SE of the regression line (Chapter 9, Section 9.3). The smaller the SE of the regression line, the smaller estimation error we can expect, which tends to happen the larger the sample, the smaller the residual standard deviation (the better the fit), the closer the target observation to the average of the x variables, and the more spread the x variables are.

The model error reflects the fact that we may not have selected the best model for our prediction. More specifically, instead of model f, there may be a better model, g, which uses the same predictor variables x or different predictor variables from the same original data. For example, when using a linear regression for prediction, we should have included interactions that we didn't, or we included interactions that we shouldn't have included. Or, maybe, the best model to predict y is not a linear regression but something else.

The last component, irreducible error, is due to the fact that we are not able to make a perfect prediction for y_j with the help of the x_j variables even if we find the best model and we can estimate it without any estimation error. Maybe, if we had data with more x variables, we could have a better prediction using those additional variables. But, with the variables we have, we can't do anything about this error. That's why its name is irreducible error.

Recall that we use the prediction interval (PI) for interval predictions, and the PI depends on the standard prediction error (SPE) (Section 13.2). It turns out that this SPE includes two of the three components of the prediction error: estimation error and irreducible error. It includes model error within the irreducible error component. It is computed using the results from a specific model, so it measures uncertainty in the context of that specific model.

For practical purposes, we shall try to reduce the estimation error and the model error, and we'll acknowledge that there is some, often large, irreducible part of the prediction error. To reduce the estimation error, we'll try to use as many observations as possible. To reduce model error, we have to find the best possible model. That is the topic of the rest of this chapter.

Review Box 13.3 Prediction error and its components

- The prediction error is the difference between the predicted value of the target variable and its actual (yet unknown) value y_j for the target observation: $e_j = \hat{y}_j - y_j$.
- The prediction error can be decomposed into three parts:
  1. Estimation error: the difference between the prediction from the estimated model and the model that would be true in the population, or general pattern, represented by the data.
  2. Model error: the difference between the prediction from the model we use and the best model.
  3. Irreducible error (idiosyncratic or genuine error): the difference between the predicted value from the best possible model and the actual value.
- The prediction interval from a linear regression includes uncertainty due to estimation error and the mix of irreducible error and model error.

13.A2 CASE STUDY – Predicting Used Car Value with Linear Regressions

Five regression models

Let's consider the predictor variables in the analysis. To model nonlinearity in age for a regression with price as the dependent variable, we may try a quadratic approximation.

Table 13.1 Versions of regression models for predicting used car price

Model No.	Used variables
Model 1	age, age squared
Model 2	age, age squared, odometer, odometer squared
Model 3	age, age squared, odometer, odometer squared, LE, condition like new, excellent condition, good condition, dealer
Model 4	age, age squared, odometer, odometer squared, LE, condition like new, excellent condition, good condition, dealer, SE, XLE, cylinder
Model 5	same as Model 4 but with all variables interacted with age

The odometer, measuring miles the car traveled (in 10 thousand mile units), is a likely strong predictor of price. We also include the more specific type of the car: LE, XLE, SE (this information is missing in about 30 percent of the observations). We also include if the ad says that the car is in good condition, excellent condition, or it is like new (this is missing for about one third of the ads). We have a binary variable capturing if the ad was posted by a dealer rather than a private person. Finally, we include whether the car's engine has 6 cylinders (20 percent of ads say this; 43

percent say 4 cylinders, and the rest have no information on this characteristic). Our data has 281 observations.

Using the data, we build various prediction models described in Table 13.1. We start by adding age and odometer reading (mileage) and then gradually add more variables.

Table 13.2 shows the results from the first four regressions. The fifth regression includes interactions with age and has many coefficients; we don't include those estimates in the table for simplicity. The table shows the regression coefficients but not the standard errors or stars for significance. This is a prediction exercise and thus statistical inference of the coefficients is not interesting – our goal is not to generalize the coefficients to the population represented by the data.

Table 13.2 Regression models for predicting used car price

Variables	(1) Model 1	(2) Model 2	(3) Model 3	(4) Model 4
age	−1,530.09	−1,149.22	−873.47	−836.64
age squared	35.05	27.65	18.21	17.63
odometer		−303.84	−779.90	−788.70
odometer squared			18.81	19.20
type: LE			28.11	−20.48
type: XLE				301.69
type: SE				1,338.79
condition: like new				558.67
condition: excellent			176.49	190.40
condition: good			293.36	321.56
cylinder_6				−370.27
dealer			572.98	822.65
Constant	18 365.45	18 860.20	19 431.89	18 963.35
Observations	281	281	281	281
R-squared	0.847	0.898	0.913	0.919

Note: Coefficient estimates for four regression models. Standard errors not shown.

Source: `used-cars` dataset Used Toyota Camry cars, Chicago area.

We can interpret the coefficients, mostly to check if the model makes sense. Age is negatively related to price, in a convex fashion, which means the left half of a U-shaped curve. The same is true for odometer reading. Cars that are advertised as being like new, in good condition, or in excellent condition, have higher prices than others.

To illustrate a point prediction, take Model 3 and values of the predictor variables for our car: age 10, odometer 12×10 thousand miles, type LE, and in excellent condition. To compute the predicted value, we use the coefficient estimates from model (3):

$$19431.89 + 10*(-873.47) + 100*(18.21) + 12*(-779.9) + 144*(18.81) + 1*(28.11) + 1*176.49 = 6072.63$$

Hence the predicted price for a car like ours is 6073 dollars.

As summarized in Table 13.3, this point prediction has the 95% prediction interval (PI) of PI[3382–8763]. Ads for cars just like ours may ask a price ranging from 3382 to 8763 dollars with a 95% chance. That's a very wide range of over 5000 dollars. The 80% prediction interval is narrower by almost 2000 dollars. Note that the first, simplest model has a point prediction of 6569 dollars, with an even wider prediction interval.

Table 13.3 Point predictions and interval predictions for a specific car, using models 1 and 3

	Model 1	Model 3
Point prediction	6569	6073
Prediction Interval (80%)	[4,296–8,843]	[4,317–7,829]
Prediction Interval (95%)	[3,085–10,053]	[3,382–8,763]

Source: `used-cars` dataset. Used Toyota Camry cars, Chicago area, N=281.

Even though our Model 3 captures a large part of the variation in prices (with an R-squared of 91.8%), there remains a large uncertainty in any given prediction. This example is instructive: despite having a great fit within the sample, predicting a single value with great confidence is hard.

If we want to sell fast or think that other features of our car make it less valuable, we may want to advertise it at a price closer to the lower end. Conversely, if we think that other features of our car make it more valuable and we have time, we may want to pick a price closer to the high end. Or, we may go for the best prediction of 6073 dollars.

13.4 The Loss Function

When making a prediction we want as small a prediction error as possible. The prediction error forms the basis of measuring the quality of our prediction. As we discussed it, it's something we can think about but not measure, as one of its components is the unknown y_j. Let's keep putting that aside, and let's think about how we can build on the prediction error to evaluate a prediction.

We certainly care about the magnitude of the prediction error: a larger error is always worse than, or at least as bad as, a smaller error. In fact, a larger error may be disproportionately worse than a smaller error. But we may also care about the direction of the error. Underpredicting the value by the same amount as overpredicting it may be equally bad, but it may be better, or it may be worse.

Ultimately, both magnitude and the direction of the error matter because predictions lead to decisions that have consequences. Thus, an error in the prediction has consequences, too. For example, when we want to predict sales of ice cream in our ice cream shop, underprediction may make us prepare less ice cream and run out of it. That's bad both because we forgo profits on that day, but also because it would hurt our reputation. If, in contrast, we overpredict sales, we may produce too much ice cream and end up with a lot of unsold ice cream. That's bad because we incur costs without revenues for unsold ice cream; ice cream is perishable, especially if it's in the display area.

To help decisions, we would attach a value to the consequences of the prediction error: a loss that we incur due to decisions we make because of a bad prediction. This idea is formulated in a **loss function**. A loss function translates the prediction errors into a number that makes more sense for the decisions and their consequences. Typically, we have more than one target observation. The loss function takes the prediction error for each of them and produces a single number. The loss function helps to rank predictions. Of course, we could simply rank the predictions based on adding up prediction errors: that would be a linear loss function. But a loss function may be a nonlinear function of the errors.

However nice it would be, specifying a loss function can be very hard in practice. It would involve quantifying all of the consequences of all of the decisions that are made based on the predictions. But some general properties of a loss function may be established without fully specifying it. Of those, the two most important are symmetry versus asymmetry, and linearity versus convexity.

Symmetric loss functions attach the same value to positive and negative errors of the same magnitude. **Asymmetric** loss functions attach different values to errors of the same magnitude if they are on different sides. Linearity versus convexity is about the magnitudes of error in quantitative predictions (or probability predictions). **Linear** loss functions attach proportionally larger loss to an error, regardless of its magnitude. In contrast, **convex** loss functions attach a disproportionately larger loss to an error that is larger.

For example, if our ice cream shop has a lot of competition, we may be more worried about damaged reputation than incurring more costs, and so our loss function would be asymmetric, with larger loss attached to a negative prediction error than to a positive prediction error of the same size. Most likely, our loss function would be convex in both directions: a larger prediction error would have disproportionately larger consequences than a smaller prediction error.

Or consider the inflation forecasts used by central banks to decide on monetary policy. The loss function of central banks should describe the social costs of wrong decisions based on erroneous predictions. The larger the prediction error, the larger the chances of wrong decisions and thus the larger the social costs. It is not straightforward to assess whether social costs are symmetric and how large they are. Arguably, however, larger errors may induce costs that are disproportionately larger: small prediction errors may not divert monetary policy from the best choice at all, or they may divert it only to a small extent that is relatively easy to reverse. Therefore, the loss function may or may not be symmetric, and it is likely to be convex.

Figure 13.1 shows some examples for the various shapes of loss functions. One loss function is linear, one is convex and symmetric, and one is convex and asymmetric, assigning smaller losses to negative errors than to positive errors.

As we discussed, ideally we would attach a loss function to the prediction error based on the actual consequences of that error in our decision situation. That's very difficult in practice, and very often it is simply impossible. Instead, we often use a generic loss function that we think represents the most important properties of our situation.

By far the most widely used generic loss function is the **squared loss**. It defines the loss as the square of the error. It is therefore a symmetric and convex loss function.

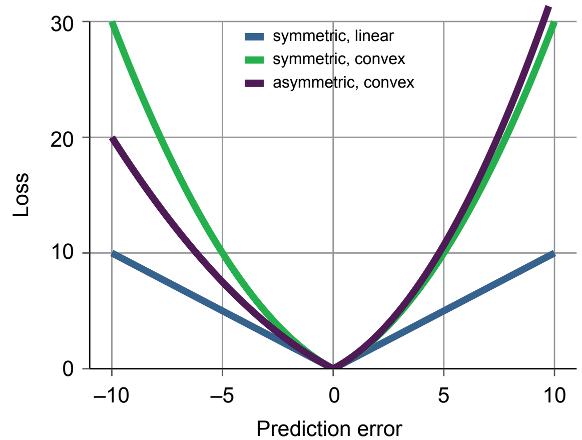

Figure 13.1 The shape of three loss functions: linear and symmetric, convex and symmetric, convex and asymmetric

Review Box 13.4 Loss functions

- A loss function translates the prediction error into losses due to such predictions.
- A loss function is necessary to evaluate the quality of a prediction.
- Two key characteristics of loss functions are symmetry and convexity.

13.5 Mean Squared Error (MSE) and Root Mean Squared Error (RMSE)

The most frequently used loss function is squared loss. When we have a prediction for multiple observations, we aggregate those squared losses across them. The most widely used aggregation is the mean, and it's called the **mean squared error**, or **MSE**. Often, we use its square root, the **root mean squared error**, or **RMSE**. Comparisons using RMSE lead to the same results as comparisons based on MSE, but the magnitude of RMSE is easier to interpret: it's the typical error we are expected to make (similar to standard deviation versus variance). In the formula, J observations are used; each is labeled with index j:

$$MSE = \frac{1}{J}\sum_{j=1}^{J}(\hat{y}_j - y_j)^2 \tag{13.6}$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{J}\sum_{j=1}^{J}(\hat{y}_j - y_j)^2} \tag{13.7}$$

The MSE formula should look familiar, because it's like the numerator of the R-squared formula for the linear regression; see Chapter 7, Section 7.9. The difference is that the R-squared is calculated using predicted values in the original data used for the regression, while the MSE as a loss function is defined for the live data, although it can be computed for the original data as well.

Review Box 13.5 Squared loss, MSE, RMSE

- The most widely used loss function is squared loss: $L(e_j) = e_j^2 = (\hat{y}_j - y_j)^2$. It's symmetric and convex.
- Mean squared error (MSE) is the average squared loss across several target observations:

$$MSE = \frac{1}{J}\sum_{j=1}^{J}(\hat{y}_j - y_j)^2$$

- Root mean squared error (RMSE) is its square root:

$$RMSE = \sqrt{MSE}$$

13.6 Bias and Variance of Predictions

Earlier, we discussed the three components of the prediction error in Section 13.3: estimation error, model error, and irreducible error. Analogously, the MSE can be decomposed into these three parts, too: estimation mean squared error, model mean squared error, and irreducible mean squared error. This decomposition is useful because it helps guide our thinking of how to find the best model: the model should give as precise estimates as possible, and it should be as close to the theoretically best model as possible. With the data we have, we can't do anything with the third component. Moreover, these two can be traded off: having a model that is close to, but not exactly, the theoretically best model may be fine if it buys us substantially smaller estimation MSE. Or, the other way around, a little more estimation error may be fine if it's more than compensated by a smaller model error.

There is another way to decompose the MSE, and that can also help thinking about the best model and potential trade-offs in finding it. That is the bias–variance decomposition.

The **bias of a prediction** is the average of its prediction error. An unbiased prediction produces zero error on average across multiple predictions. A biased prediction produces nonzero error on average; the bias can be positive or negative.

The **variance of a prediction** describes how it varies around its average value when multiple predictions are made. It's the variance of the prediction error around its average value. That average value can be zero or not, so the prediction may be unbiased or biased. The variance is zero if the prediction error is the same for all predictions. The variance is higher the larger the spread of specific predictions around the average prediction.

It turns out that the MSE is the sum of the squared bias and the prediction variance:

$$\begin{aligned} MSE &= \frac{1}{J}\sum_{j=1}^{J}(\hat{y}_j - y_j)^2 \\ &= \left(\frac{1}{J}\sum_{j=1}^{J}(\hat{y}_j - y_j)\right)^2 + \frac{1}{J}\sum_{j=1}^{J}(y_j - \bar{\hat{y}})^2 \\ &= Bias^2 + PredictionVariance \end{aligned} \tag{13.8}$$

This decomposition helps appreciate a trade-off: a biased prediction with small variance may be better than an unbiased prediction with a large variance. Of course, a prediction with both smaller bias and smaller variance is always better.

A metaphor that is often invoked in this context is target shooting. Unbiased shooters hit the target on average. But they can do that in various ways: they can be very close to the target all the time; or they can shoot over a very wide range, but the errors average to zero. The first example is one of low variance, the second is one of high variance. Moreover, when comparing an unbiased but high-variance shooter with a biased but low-variance shooter, we may prefer the latter. That's because the first one is prone to make big mistakes, while the second one is known to make small mistakes, even if those mistakes don't average to zero.

The linear regression, estimated by OLS, gives an unbiased prediction. Thus, the ranking of linear regression models by their MSE means ranking them by their prediction variance. But there are other models that are a little biased but produce a smaller prediction variance and achieve a better MSE that way.

13.7 The Task of Finding the Best Model

After introducing the theoretical basis for comparing the predictions of various models, we can now discuss how to do that in practice. The fundamental difficulty stems from the fact that what we have is the original data. Thus, we can compare the predictions of various models using the original data only. However, what we want is the best prediction in terms of the live data, not the original data.

An important dimension along which various models may differ is **model complexity**. For example, in a linear regression, a more complex model has more coefficients. That may be because it has more right-hand-side variables. Of two linear regressions that make use of the same x variables in the data, a more complex model has more coefficients because of more complicated functional forms (more variables squared terms, higher-order polynomial terms, more knots for splines, and so on), or more interactions. In our case study of predicting used car value, the five models we considered were five linear regressions of increasing level of complexity. Finding the best model involves finding the right degree of complexity. That turns out to be a balancing act.

Using the language we introduced in the context of regression (Chapter 7, Section 7.9), the best model is the one that fits the live data best. But there is no way to tell how predictions would fit the live data. Instead, we have to choose the best model using the original data. The fit of a regression model to the original data can be measured by the R-squared. Recall that the R-squared is just like the MSE, except it is computed using predictions within the original data, and it is divided by the variance of y and subtracted from one:

$$R^2 = 1 - \frac{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{13.9}$$

So we can rank models by their R-squared in the original data. This ranking is the same as ranking by their MSE would be if it were computed using the original data.

Unfortunately, though, the best R-squared in the original data may not give the best MSE in the live data. In fact, it never does. That's because a more complex model always gives a better R-squared. But that complexity may reflect patterns that are specific to the original data but would not be there in the live data.

In fact, comparing two models, Model 1 can give a worse fit in the live data than Model 2 in two ways. First, Model 1 may give a worse fit both in the original data and the live data. In this case, we say that Model 1 **underfits** the original data. This is quite straightforward. Second, however, Model 1 may actually give a better fit in the original data, but a worse fit in the live data. In this case, we say that Model 1 **overfits** the original data. This is less straightforward, but it's just as important.

Overfitting is a serious threat, because all we have is the original data to find the best model. The possibility of overfitting the original data means that the model that gives the best fit in the original data may not be the one that gives the best prediction in the live data. A more complex model always gives a higher R-squared in the original data. That's because a more complex model tends to find more detailed patterns in the original data. But some of those more detailed patterns will be there for the original data only; they will not be present in the live data. Thus, a very complex model would produce a very good fit in the original data, but, with a high chance, it would in fact overfit the original data.

Consider our example of predicting sales in our ice cream shop. Suppose that we have daily time series data on temperature and sales for three years, with about 1000 observations in total. Model 1 may be a linear regression with month dummies (11 variables), day of the week dummies (6 variables), and temperature in a quadratic function. These are 20 coefficients, including the intercept. Model 2 may add interactions of month with temperature (in quadratic). These are 22 additional variables, with a total of 42 coefficients to estimate. Model 3 includes interactions of month, day of week, and temperature. This means 12×7 month–day interactions (minus the reference group), twice as many temperature coefficients (in quadratic), plus day of the week–temperature interactions for the reference month category, with close to 300 coefficients to estimate in total.

In this example, Model 1 may underfit the original data compared to Model 2: the temperature–sales pattern may differ by months both in the original data and the live data, which is captured in Model 2 but not in Model 1. In contrast, Model 3 may overfit the original data. One way to understand Model 3 is that it fits a regression of sales and temperature in a quadratic form for each month and day of the week separately. Consider Wednesdays in August, with around 12 to 15 such observations in the data (three years means three Augusts, each with 4 or 5 Wednesdays). That's a very small sample. We fit a quadratic in temperature on this data. A lot of other things affect sales that may have happened on August Wednesdays during these three years that won't necessarily happen in the live data (one of our competitors ran a price discount, one of our refrigerators was out for a few hours, and so on). And some of those other things may have happened on days with unusually high or unusually low temperatures. This is a small sample, so one or two such events could tilt the regression curve to a large degree in some way. But that's not how the sales–temperature pattern will play out in the live data. Our very complex model is bound to fit some of these patterns in the original data. But that's overfitting.

Figure 13.2 illustrates underfitting and overfitting and its relation to model complexity. It shows many hypothetical models. The horizontal axis shows their complexity: more complex models are to the right. The vertical axis shows the RMSE of each model: higher values show worse fit. The blue curve shows RMSE in the original data; the green curve shows RMSE in the live data. Models with very low complexity give a bad fit to both the original data and the live data. Looking at more and more complex models, the fit in the original data keeps getting better and better. However, the same is not true for fit in the live data: after a point, fit starts to become worse as model complexity increases. Too complex models overfit the original data: they tend to give much worse fits in the live data than in the original data.

Finding the best model requires using the original data to infer something about the live data. Similarly to what we did when we discussed inference earlier in Chapter 5, it makes sense to split this problem into two. The first problem is the original data versus the population, or general pattern, it represents. The second problem is the two worlds: the population, or general pattern, that is relevant

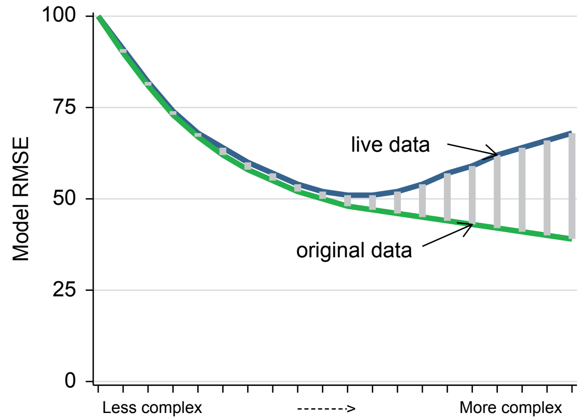

Figure 13.2 Underfitting and overfitting the original data

Note: RMSE on artificial original and live data; the grey bars indicate the difference.

for the original data, and the other world that will be relevant for the live data. Thus, the problem of overfitting the original data is best split into two problems, too: fitting patterns in the original data that are not there in the population, or general pattern, it represents; and fitting patterns in the world of the original data that will not be there in the world of the live data.

In the following three sections we'll introduce three methods that can help with the first problem: selecting the model that is the best in terms of prediction in the general pattern represented by the original data. We'll turn to the second problem, external validity, afterwards.

Review Box 13.6 Main concepts of prediction

- The best model gives the best prediction in the live data.
- The only data we can use to find the best model is the original data.
- The model that gives the best prediction (best fit) in the original data may not give the best prediction (best fit) in the live data.
- Overfitting means achieving a good fit in the original data by finding patterns that are specific to the original data only and would not achieve a good fit in the live data.

13.8 Finding the Best Model by Best Fit and Penalty: The BIC

One solution to finding the best model is using all of the original data for estimation and finding the model that fits the original data best, but measuring fit in a way that discourages overfitting. In effect, that means using a measure of fit that has a **penalty for model complexity**. In essence, a measure of fit with a penalty term has two parts: one part measuring fit, and another part measuring complexity. The final measure is better, the better the fit if the complexity is the same, or if the complexity is less but the fit is the same. Comparing models with different fits and different complexities at the same time, the two properties are traded off to rank the models.

There are several measures of fit that include a penalty term for model complexity. They differ in how they trade off fit with complexity. We mention three measures: **adjusted R-squared**, **AIC**

(Akaike's information criterion), and **BIC** (the Bayesian information criterion). Most statistical software shows these measures by default when reporting the results of regression models.

Of the three, we suggest using BIC because many data analysts have found it to be the best of these similar measures. It is also called the Schwarz criterion. The BIC is a statistic measuring how far the predicted values are from actual values so the lower the BIC value, the better the expected fit. Thus, a lower BIC implies a better fit, a less complex model, or both.

It turns out that the BIC is derived from a set of complicated assumptions regarding the data. Most of those assumptions are hard, or impossible, to check. For that reason, the BIC is not necessarily an ideal way to select the model that would provide the best fit to the live data. In particular, the BIC is sometimes too harsh on model complexity, giving a lower ranking to models that are more complex even when those end up being better at prediction in the live data.

Despite its shortcomings, in many cases the BIC can help avoid severe overfitting and pick the best model, or at least one of the best of models. It is easy to calculate, and it often gives a good enough answer.

The most important caveat to keep in mind is that the BIC is calculated using the original data. If external validity is high and the patterns in the live data are likely similar to the patterns in the original data, that's fine. However, if external validity is low, we have a problem. The BIC is not tuned to appreciate whether external validity is high or low. It is tuned to measure fit in the original data and penalize complexity.

Review Box 13.7 The BIC

- The BIC is a measure of the fit of a model using all the original data.
- The BIC penalizes model complexity and thus helps avoid overfitting the original data.

13.9 Finding the Best Model by Training and Test Samples

An alternative to using all the original data and a measure of fit that penalizes model complexity is imitating the original versus live data problem in a more direct way. All we have is the original data. But we can split the original data into two, and use one part for estimation and the other part for measuring fit.

The first part of the data is called the **training set**, or **training sample**. We use this part of the data as if it were all the original data we had. In particular, we do exploratory data analysis that we use to specify regression models, and then we estimate the regression models using the training set only.

The second part of the data is called the **test set** or **test sample**. This is the part we use to evaluate the predictions. In particular, we pretend that we can observe x in this data but not y, and we use the model estimates from the training set to predict $\hat{y}$ for each observation in the test set. Then, we compare those predictions with the actual y values to calculate the prediction error and then the measure of fit, such as MSE, RMSE, or something else if we have a different loss function. The model that gives the best fit to the test set is the best model.

Finally, once the best model is found, we re-estimate that model using the entire original data. We use this estimated model for prediction.

Let's spend a little more time on related language that is used broadly in predictive analytics. In the prediction setup, **training a model** means all the steps of data analysis that lead to a prediction:

selecting the x variables, maybe transforming the y and/or some of the x variables, specifying the model, and estimating it. In predictive analytics, **testing** means predicting y values using a model (that was "trained" previously) and evaluating that prediction. Importantly, this kind of testing is a different thing from testing hypotheses that we discussed in Chapter 6. The method itself is called the **train–test method**, or the **training–test split**, to evaluate predictions.

All of this sounds quite intuitive, indeed: the proof of the pudding is in the eating. If we want to find a model that best fits data that is similar to, but not the same as, the data used for the estimation, let's do just that: let's find the best fit on data that is similar to, but not the same as, the data we use for estimation. However, there are some details that we need to decide on, and those details may matter. And, as we'll see, there are issues with this approach, no matter how we decide on those details.

The details to be decided for the training–test split are how many observations to allocate to the training set and the test set and how to select those observations. In general, data analysts choose more observations to be in the training set than the test set. Typical choices for the size of the training sample are 80% or 90% of the full original data, with corresponding test sample sizes of 20% and 10%.

How to select the observations is the other decision to make. The guiding principle is to have the test set as similar to the live data as possible. Unfortunately, however, we know little about what the live data will look like. In particular, we don't know whether and how the patterns of association between predictors and the outcome will differ from the original data. For this reason, the usual choice is a random sample. The training set is a random subsample of the original data; the test set is the remaining part that is thus also a random sample.

Three issues makes the training–test split a less-than-ideal method for choosing the best prediction method.

First, this method uses fewer observations for model estimation than are available, at least when trying to find the best model. The training set is a subsample of the original data. Recall that the prediction error tends to be larger when the estimation error is larger, and the estimation error is larger the fewer observations are used. This actually is the reason why, once the best model is found, we go back and use all observations in the original data to estimate the model for the actual prediction. So we avoid the issue for the prediction itself. But higher estimation error may still matter for our ability to find the best model. We try to minimize the issue by having the training sample contain more observations than the test sample, such as 80% versus 20%.

The second of the three issues is that the training–test method doesn't really avoid overfitting the original data. True, it avoids overfitting the training sample, as the predictions are evaluated without using observations from that sample. But it may very well overfit the test sample. In fact, the test sample may have patterns in it that are specific to it and not present in the live data, just as the training sample would have such patterns. Thus, overfitting the test sample is a likely problem.

The third issue is common across the training–test method and the BIC and the similar measures. It's the fundamental issue with external validity. All we have is the original data. Thus, with a clever method, we may be able to find the best model for the population, or general pattern, represented by the original data. That way we can avoid overfitting patterns that are there in the original data but would not be there in the population represented by the original data. Whether the model that we find this way is also the best model for the live data requires a leap of faith: external validity. As we'll see later, we can sometimes find some guidance for that leap even in the original data, but that requires thinking instead of using a method off the shelves.

Before turning to external validity, we discuss an improvement on the train–test method, which helps a lot with the second issue we listed (overfitting the test sample).

13.10 Finding the Best Model by Cross-Validation

Using a test set to evaluate predictions rewards models that fit patterns that are in the test set and penalizes models that fit other patterns which are not in the test set. This way, the procedure avoids overfitting in the training data. At the same time, we may fit a model on patterns that are specific to the test set. An improved version of splitting the data is called k-fold cross-validation, and it remedies this problem.

Cross-validation (**CV**), also known by a more precise name, **k-fold cross-validation**, is a method of sample splitting that involves taking more than one split of the original data into training and test sets, and then looking at the average fit across all test sets. Each observation will appear exactly once in a test set and $k-1$ times in the training sets. A training set–test set combination is called a **fold**. A k-fold cross-validation works with k folds.

For example, with 10-fold cross-validation, we split the original data into ten subsets in a random way, so each subset consists of 10% of the observations. Then we take each of the ten 10% samples as a test set and define the corresponding training set as the other 90%. There are ten folds: ten training–test combinations. The test sets do not overlap with each other, and they cover all observations in the original data. The training–test method described above is hence carried out ten times. We can calculate the loss function, such as the MSE, ten times: $MSE_1 \ldots MSE_k$. The overall fit of the model is measured by the average of the ten MSE values, or the RMSE, which is the square root of this average:

$$MSE_{cv(k)} = \frac{1}{k}\sum_{i=1}^{k} MSE_i \tag{13.10}$$

$$RMSE_{cv(k)} = \sqrt{\frac{1}{k}\sum_{i=1}^{k} MSE_i} \tag{13.11}$$

This approach decreases the role of a particular test set in determining the MSE and thus the risk of overfitting the patterns of a particular test set.

A further improvement would repeat cross-validation many times, each for a different random partition of the dataset, and compute the overall average MSE. This approach further decreases the problem of overfitting a particular test set by selecting a larger number of different test sets. In most cases, this refinement offers little further improvement.

Thus, k-fold cross-validation helps greatly with what we listed as the second issue with the training–test method: overfitting the test sample. Thus, k-fold cross-validation is a better method to find the best model than a single training–test split.

In addition, k-fold cross-validation has a clear advantage over BIC, too. While BIC can be applied to regression models, k-fold cross-validation is a method that can be applied to any prediction model. As a result, k-fold cross-validation is the most widely used method to find the best model in predictive analytics. It requires a lot of work, but that work is done by the computer, not the analyst. k-fold cross-validation is a built-in feature of most statistical software.

Review Box 13.8 Training–test split and k-fold cross-validation

- The training–test split partitions the original data into two sets in a random way. All analysis and estimation takes place using observations in the training set. The evaluation of prediction takes place using observations in the test set.
- The training–test split avoids overfitting the training set; however, it may overfit the test set.
- k-fold cross-validation improves on the single training–test split by repeating it k times. Each split is called a fold. The prediction of a model is evaluated across the folds, such as the average MSE.
- k-fold cross-validation is a good way to find the model that would give the best prediction for the population, or general pattern, represented by the original data.

13.A3 CASE STUDY – Predicting Used Car Value with Linear Regressions

Finding the best model

Let us go back to our case study and find the best of the five predictive models using the tools we have just covered.

Table 13.4 Car price models estimated using all original data and measures of fit using all original data

Model	N vars	N coeff	R-squared	RMSE	BIC
Model 1	1	3	0.85	1755	5018
Model 2	2	5	0.90	1433	4910
Model 3	5	9	0.91	1322	4893
Model 4	6	12	0.92	1273	4894
Model 5	6	22	0.92	1239	4935

Note: Five linear regression models; see Table 13.1 for details. Here N vars is the number of variables; and N coeff is the number of all coefficients including the intercept.

Source: `used-cars` dataset. Used Toyota Camry cars, Chicago area, N=281.

The loss function for predicting the value of our used car may or may not be symmetric, depending, among other things, on how we value money and how we value how much time it takes to sell our car. Too low a prediction may lead to selling our car cheap but fast; too high a prediction may make us wait a long time and, possibly, require us to revise the sales price downwards before selling our car. The loss function may be linear or convex, again, depending on how we value money and time. We will consider a symmetric and convex loss function, the one captured by RMSE.

We have 281 observations, and our regression models have few variables, but the most complex models include many interactions. For several categories such as car types, we have a few dozen observations driving results. Hence, we may indeed overfit the data, especially with models that include many interactions. To find the best predictive model, we first look at BIC values along with R-squared and the RMSE in Table 13.4.

Second, we cross-validate using k-fold cross-validation. We set $k = 4$, given the small sample (a few hundred observations is a small sample for prediction). This means splitting the data into four in a random fashion to define the four test sets. Thus, each fold has 1/4 of the observations in the test set and 3/4 of the observations in the training set. For each fold, we estimate ("train") all five models using the training set and calculate MSE on the corresponding test sample. We then average out MSE values over the four test samples for each model. The results are shown in Table 13.5.

Table 13.5 Car price models estimated and evaluated using 4-fold cross-validation and RMSE

Fold No.	Model 1	Model 2	Model 3	Model 4	Model 5
Fold1	1734	1428	1331	1395	1391
Fold2	2010	1781	1692	1638	1693
Fold3	1465	1251	1256	1253	1436
Fold4	1823	1325	1250	1246	1307
Average	1769	1460	1394	1392	1464

Note: Five linear regression models; for details see Table 13.1. Cross-validated RMSE values.

Source: `used-cars` dataset. Used Toyota Camry cars, Chicago area, N=281.

Both approaches suggest that Model 3 and Model 4 are the ones that have the best prediction properties. They have the lowest BIC (4893 and 4894), and they have the lowest average cross-validated RMSE (1394 and 1392). The ranking of Model 3 and Model 4 is different, but the actual difference between the two models is very small in terms of both of the performance measures. So here BIC and cross-validation found the same two models to be the best. From our experience, that turns out to be a common outcome when we compare regression models.

As a final point on model selection in the case of a range of less and more complicated models, note that choosing a simple model can be valuable as it may help us avoid overfitting the live data. Whenever simple and complicated models have RMSE values that are very close, we suggest picking a simple one. Here that implies choosing Model 3.

13.11 External Validity and Stable Patterns

With the help of the BIC or k-fold cross-validation, we can find the model that is best in predicting *y* in the population, or general pattern, represented by the original data. But that's just the first of the two steps of finding the model that will give the best prediction in the live data. We now turn to external validity, the second step and perhaps the biggest – certainly the hardest – of all problems with prediction.

In the abstract, external validity is about whether, and to what extent, we can generalize findings in the population, or general pattern, represented by our data to the population, or general pattern, we are truly interested in (Chapter 5, Section 5.8). In predictive analytics, external validity means the extent to which the prediction errors are similar in the live data and in the population, or general patterns, represented by the original data. Thus the external validity of a prediction is high if the model that produces the best prediction in the population, or general pattern, represented by the original data is also the model that produces the best prediction in the population, or general pattern, that is behind the live data. Moreover, external validity of a prediction is high if the fit (e.g., RMSE) is similar in the two.

For high external validity, we require that the patterns of association between y and x are similar in the two worlds: the population, or general pattern, behind the original data, and the population, or general pattern, behind the live data. All models make use of those patterns of association to make a prediction. Thus, for a model to be best in both worlds, those patterns of association must be similar in the two worlds. This requirement is one of **stability**: the patterns of association between y and x should be stable across the two worlds. Because the original data is from the past and prediction is usually about the future, this also requires **stationarity**, which is a statistical concept describing stability across time (Chapter 12, Section 12.3). But stability may require more than that. Recall from Chapter 5, Section 5.8, that other frequent threats to external validity are differences across space or groups of observations. We need stability along those dimensions, too.

In our ice cream shop example the main issue is stability over time. We have data from three recent years, and we uncovered the patterns of association between sales, temperature, and monthly and daily seasonality in the data. The question of external validity is to what extent those patterns are going to be the same in the near future we want to predict. We may worry about changes in demand for many reasons, including changes in the structure of competition, changes in income and tastes of potential customers, or changes in the size of the customer pool.

Or take our other example, predicting default of people on personal loans. Suppose that our data covers applicants from one geographic region. However, the way the default probability is related to personal characteristics may be different in other regions. For example, some industries may be expanding in our region but contracting in other regions, thus employment in those industries is associated with lower loan default in our region but less so in the other regions. That leads to low external validity of a prediction that uses the industry of employment as a predictor variable.

Being able to list our worries is fine, but how can we tell whether they make external validity of a prediction high or low? And can we do anything to select a model with higher external validity?

Here is where **domain knowledge** and thinking can help. Domain knowledge means knowledge of the mechanisms behind the patterns of association of y and x. For example, the best-predicting model we find may make use of some patterns of association that are surprising. Then we need to think about whether those associations are likely true in the live data, too, or if they are specific to the original data. Often, surprising patterns are surprising because they are not generally true.

When external validity is not perfect, we may want to reflect that in our prediction in some way. Intuitively, less-than-perfect external validity adds to the uncertainty of our prediction. Recall that one way to represent that uncertainty is giving an interval prediction instead of, or complementing, a point prediction. Worries about external validity could be reflected in a wider prediction interval for the same level of confidence or by attaching a lower level of confidence to a prediction interval. Unfortunately, it's very hard to quantify external validity or lack of it, so these modifications are most often ad-hoc.

In our ice cream example, we may know of a few things that should affect demand. Four new large office buildings just opened within a few blocks, increasing the pool of potential customers. At the same time, a new ice cream shop opened further down the road, increasing competition. Doing some math, we calculate that the two may just balance out each other, so, as a result, we may face the same amount of demand. Thus, we may conclude that the predictions from our data have high external validity for the near future. But those developments increase uncertainty. Thus, we should probably attach more uncertainty to our predictions, either in a formal way, by making the prediction interval wider, or by making business decisions with the consideration that our prediction may be off by a wider margin than what the prediction interval suggests.

Analyzing our original data can also help. For example, when the original data is from the same geographic area, covering the same groups of observations, as what the live data will be, we have more reason to assume stability. Or, if our data covers many geographic areas, we can check if the patterns of association are similar across them. Similarly, we can check stability in time if our original data is from different points in time. Of course, all we can find is evidence for stability in the past, which doesn't guarantee stability in the future. But patterns of associations that were stable in the past are more likely to remain stable in the future, too. As a flipside, if we find that the patterns of association changed substantially from year to year in the past, we have less hope of using historic data to produce good predictions for the future.

Review Box 13.9 External validity of a prediction

- A prediction has high external validity if the best model for the population, or general pattern, behind the original data is also the best model for the population, or general pattern, behind the live data.
- High external validity of a prediction requires that the patterns of association between y and x are stable, so that they are very similar in the original data and the live data.
- Domain knowledge, thinking, and data from multiple time periods can help assess external validity.

13.A4 CASE STUDY – Predicting Used Car Value with Linear Regressions

External validity

This case study offered our first insight into prediction. We started with data cleaning, and we specified five different linear regressions to predict used car prices. To find the best of the five models we considered two methods: using all data to estimate the models and calculate their BIC, and using 4-fold cross-validation to calculate cross-validated RMSE. In the end, both methods suggested that we choose Model 3 or 4, which had many, but not all, of the predictor variables in the data, and which didn't have any interactions. More complex models were found to overfit the data. Out of Models 3 and 4 with equal fit, we pick the simpler one, Model 3.

We motivated this exercise by focusing on a single car with specific characteristics. Yet the model selection criteria we used looked at all predictions, including cars that were different from that car in terms of age, odometer reading, and other characteristics. We had two reasons to do so. First, we wanted a model that is expected to perform well. Looking at the average of predictions (or squared prediction errors) across many cars was one way to assess that expectation. Second, another motivation for the analysis was to see what price we could predict if we were to wait a year or two and sell a car with higher age, more miles, and so on. Thus we wanted a model that produces good predictions not only for the particular car we have but for the cars with other characteristics, too.

Using Model 3, our point prediction for the type of car we want to sell was 6073 dollars with a wide 80% prediction interval, of 4317 to 7829 dollars.

How confident can we be that in the case we need to predict the price of a similar car, we can rely on this estimate? Estimation error and irreducible error make the PI wide, reflecting a high level of uncertainty. Still, these results can help us advertise our car. If we want to sell quickly and know of some unmeasured negative features about our car, we may go towards the lower end of the interval. If our car is in a truly excellent condition for its age and miles and we have time, we may go for the upper end.

Finally, let's think about external validity. Shall we expect the patterns of association between y and x to be the same in the original data (from 2018) and the live data (now)? The data is from the same geographic area, covering very similar kinds of cars. Thus, what we have to worry about is stability in time. Are car prices related to age, mileage, and car type the same way now as in 2018? To evaluate that, we need to think about inflation, changes in income conditions in the area, and perhaps changes in the demand for combustion-engined cars. If we are close to 2018, then with all likelihood these factors will have changed little. As we move further away from 2018, external validity likely decreases. We could reflect this by making the PI even wider. Or, we'd perhaps be better to collect new data from the more recent past to carry out a predictive analysis.

From a methodological point of view, two main conclusions emerge. First, the best prediction model was the one with a medium level of complexity. Some models were too simple and led to underfitting the data; other models were too complex and led to overfitting the data. Second, we have used two ways to select the best prediction model: using all data and BIC, and using cross-validation. Here, the two methods ended up selecting the same two models that have very similar performance. In such a case, we argued that the best choice is to pick the simpler model.

13.12 Machine Learning and the Role of Algorithms

We close this chapter with a note on terminology often used both by data analysts and the general public. As we introduced earlier, the term **predictive analytics** is often used for data analysis whose goal is prediction. But a more popular, and related, term is **machine learning**.

In essence, **machine learning** is an umbrella concept for methods that use algorithms to find patterns in data and use them for prediction purposes.

In the context of predictive analytics, an **algorithm** is a set of rules and steps that defines how to generate an output (predicted values) using various inputs (variables, observations in the original data). A **formula** is an example of an algorithm – one that can be formulated in terms of an equation. For example, the OLS formula for estimating the coefficients of a linear regression is an algorithm. It's an equation that specifies the rules to follow to arrive at the OLS estimate of a coefficient using observations and variables in actual data.

But not all algorithms can be translated into a formula. The bootstrap estimation of a standard error (Chapter 5, Section 5.6) is an example. It starts with random sampling, continues with estimation, repeating this across all bootstrap samples, storing the estimates from each bootstrap sample, and then combining those to calculate the standard error. Here the estimation part may be translated into a single formula, and how we combine the various estimates may have a formula, too (the standard deviation). But other steps are verbal descriptions without a mathematical formula. Yet another example is k-fold cross-validation. It starts with creating the k random partitions of the original data, followed by all the steps of data analysis to get an estimate of a model using the training set, then using the test set to carry out and evaluate the prediction, repeating this for all k folds, and then combining these evaluations across each fold.

An important aspect of the algorithms in machine learning methods is their heavy use of machines, that is computers. Computers are used to plug data into formulae. But computers are used for all other parts of the algorithm, too. The main feature of algorithms is that they specify each and every step to follow in a clear way. Those steps can therefore be translated into computer code and make the computer follow those steps.

Besides machines, the other important part of machine learning is, well, learning. Algorithms in machine learning are used to learn something from the data. We may use the data and an algorithm to learn what the predicted value of y is if we combine the x variables using a particular model. But, in a narrower sense, learning here means learning which model is best for predicting y as well as what that predicted value is.

Not all data analysts agree on what methods constitute machine learning and what methods don't. Calculating the mean of y in the data and using that mean to predict a future value $\hat{y}_j$ can be viewed as machine learning, but many data analysts wouldn't call it that. We use the computer and an algorithm (a formula) to learn something about the future predicted value. But what we learn is the predicted value only, not whether it's a prediction that is better than some other prediction. It's perhaps too simple, too, to qualify for such a fancy name. In contrast, many data analysts would be comfortable using the term machine learning for using 10-fold cross-validation to compare that prediction to another one.

In a sense, machine learning is an approach to predictive data analysis. This approach is derived from the focus on achieving the best possible prediction from available data. That has two broad implications.

First, understanding the patterns of associations between y and x is of secondary importance in machine learning. To be more precise, these patterns are important in the sense that they should be stable across time and other dimensions that make the original data and the live data differ from each other. But beyond that stability, why y and x are associated with each other is not interesting for making a prediction.

Second, the machine learning attitude includes a preference for evaluating methods based on data as opposed to abstract principles. That's because we want to make a prediction here and now, from the original data we have to the live data we'll face. We don't want to devise a prediction rule that will be fine for every possible scenario with potentially different original data or live data.

One example of the preference for data-driven methods is finding the best prediction model using cross-validation. In principle, the BIC can be used for the same purpose for an estimated model using the entire original data. But the BIC is derived as a best measure of fit if some assumptions are true about the data. Thus, using cross-validation better conforms to the machine learning attitude. In the end, the BIC and 10-fold cross-validation may lead to the same conclusion, as they did in our case study. When they don't, the machine learning approach trusts cross-validation.